

PRODUCT REQUIREMENTS DOCUMENT

FraudShield Agent

AI-Powered Real-Time Fraud Detection with Explainable Alerts

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CONFIDENTIAL — PORTFOLIO PROJECT

1. Executive Summary

Problem Statement: Fraud operations analysts at fintech companies are overwhelmed by high volumes of transaction alerts, the majority of which are false positives. Existing rule-based systems flag transactions without explaining why — forcing analysts to manually investigate each alert, slowing response time, and increasing operational cost. Missed fraud events carry serious financial and reputational consequences.

FraudShield Agent is an AI-powered fraud detection system that classifies credit card transactions in real-time using an XGBoost classification model, and translates model outputs into plain-English explanations via an LLM layer. It is designed for internal fraud operations analysts as the primary user persona.

The core PM challenge is the Precision vs. Recall tradeoff: optimizing to minimize false positives (flagging legitimate transactions) while maintaining adequate recall (catching actual fraud). This PRD defines the product vision, user stories, success metrics, feature prioritization, and risk mitigation strategy.

2. Product Vision & Goals

2.1 Vision Statement

Empower fraud operations analysts with an AI co-pilot that detects fraud faster, explains decisions in plain English, and continuously improves through analyst feedback — reducing investigation time by 40% while maintaining a false positive rate under 5%.

2.2 Business Goals

- Reduce average fraud investigation time from ~8 minutes to under 5 minutes per alert
- Achieve false positive rate $\leq 5\%$ (Precision ≥ 0.95) within 6 months of launch
- Maintain fraud recall ≥ 0.85 (catch at least 85% of actual fraud)
- Decrease analyst escalation rate by 30% through self-explanatory alert context
- Generate a feedback loop to retrain the model monthly using analyst decisions

3. User Personas

PRIMARY PERSONA	SECONDARY PERSONA
Riya — Fraud Ops Analyst <i>Senior Fraud Analyst, mid-size fintech</i> Goals: • Quickly triage high volumes of fraud alerts • Understand why a transaction was flagged • Minimize false positives impacting customers	Arjun — Risk & Compliance Manager <i>Risk Manager, reports to VP of Operations</i> Goals: • Monitor fraud detection performance across team

Pain Points:

- "Black box" alerts with no explanation
- High alert volume, low signal-to-noise ratio
- Manual cross-referencing across multiple tools

- Ensure regulatory audit trail for flagged decisions
- Reduce financial exposure from undetected fraud

Pain Points:

- No visibility into why alerts are escalated/resolved
- Reporting is manual and time-consuming
- Difficult to assess model performance over time

4. Problem Framing — Precision vs. Recall

The fundamental PM challenge in fraud detection is managing the tradeoff between two competing success conditions:

HIGH PRECISION**Minimize False Positives**

When we flag a transaction, we're usually right. Fewer legitimate transactions are blocked — improving customer experience and analyst efficiency.

Risk: May miss some actual fraud (lower Recall)

HIGH RECALL**Minimize False Negatives**

Catch as much fraud as possible. Fewer fraudulent transactions slip through — reducing financial loss and regulatory exposure.

Risk: More false alerts overwhelm analysts (lower Precision)

FraudShield Agent Decision Primary optimization target: Precision ≥ 0.95 (false positive rate $\leq 5\%$) Secondary constraint: Recall ≥ 0.85 (catch at least 85% of fraud events)
Rationale: In B2C fintech, false positives directly harm customer experience and erode trust. A missed fraud event is recoverable; blocking a customer's valid transaction causes churn.

5. User Stories

Epic 1: Real-Time Transaction Classification

- US-01: As a fraud analyst, I want transactions to be automatically scored in real-time so that I only see alerts above a defined risk threshold.
- US-02: As a fraud analyst, I want each flagged transaction to show a confidence score (0–100) so that I can quickly prioritize the highest-risk alerts.
- US-03: As a fraud analyst, I want to see a plain-English explanation of why a transaction was flagged (e.g., 'Transaction amount is 3x the user's 30-day average, originating from a new device') so that I can make faster, informed decisions.

Epic 2: Analyst Workflow Integration

- US-04: As a fraud analyst, I want to approve or reject an alert with a single click, and add optional notes, so that my decision is logged for model retraining.

- US-05: As a fraud analyst, I want to filter the alert queue by risk score, transaction type, or time range so that I can focus on the most critical cases first.
- US-06: As a fraud analyst, I want to view a user's transaction history timeline alongside a flagged transaction so that I can assess behavioral deviation without switching tools.

Epic 3: Model Explainability

- US-07: As a fraud analyst, I want to see the top 3 contributing features for each flagged transaction (e.g., amount deviation, merchant category, geographic anomaly) so that I understand the model's reasoning.
- US-08: As a fraud analyst, I want to see a confusion matrix summary in the dashboard so that I can assess overall model performance at a glance.

Epic 4: Manager & Audit

- US-09: As a risk manager, I want a daily performance report (precision, recall, alert volume, resolution rate) so that I can track system health without querying raw data.
- US-10: As a risk manager, I want all analyst decisions to be timestamped and exportable so that I can produce regulatory audit reports on demand.

6. Feature Prioritization — RICE Framework

RICE Score = (Reach × Impact × Confidence) / Effort. Scores above 100 are P1 (High). 50–100 are P2 (Medium). Below 50 are P3 (Low).

Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority
Real-time transaction scoring	10,000	3	90%	3	9,000	P1 — High
Plain-English alert explanations (LLM)	10,000	3	85%	4	6,375	P1 — High
One-click analyst decision logging	8,000	2	90%	2	7,200	P1 — High
Top-3 feature contribution view	8,000	2	80%	3	4,267	P1 — High
Alert queue filters	7,000	2	85%	2	5,950	P1 — High
User transaction history timeline	6,000	2	75%	3	3,000	P2 — Medium
Confusion matrix dashboard widget	5,000	2	80%	2	4,000	P2 — Medium
Daily performance email report	3,000	1	90%	2	1,350	P2 — Medium
Monthly model retraining pipeline	10,000	3	70%	5	4,200	P2 — Medium

Regulatory audit export (CSV/PDF)	2,000	2	85%	3	1,133	P3 — Low
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7. Functional Requirements

7.1 Classification Engine

- System shall score each incoming transaction within 300ms using a pre-trained XGBoost model
- Model trained on European Credit Card Fraud Dataset (Kaggle); updated monthly via analyst feedback
- Risk score output: 0–100 scale; transactions ≥ 70 trigger an analyst alert
- SMOTE oversampling applied during training to address class imbalance

7.2 Explainability Layer (LLM)

- Each flagged transaction shall generate a 2–3 sentence plain-English explanation via LLM (Claude / GPT-4o)
- Explanation must reference at least 2 contributing features identified by SHAP values
- Explanation shall be generated within 1 second of score computation
- LLM prompts shall be structured to avoid hallucination — grounded in transaction metadata only

7.3 Analyst Interface

- Alert queue displayed in descending risk score order by default
- Filters: date/time range, transaction amount range, merchant category, risk score band
- Per-alert actions: Confirm Fraud / Mark Legitimate / Escalate, with optional free-text notes
- Transaction history timeline: last 90 days, same user, highlighting statistical outliers

8. Success Metrics

The success matrix is structured across two layers: Model Metrics track ML performance and data quality; Product Metrics track user behaviour and business outcomes. Both layers must be healthy for FraudShield to be considered successful.

Metric	Baseline	Target (6 mo)	How to Measure	Why It Matters
LAYER 1 — MODEL METRICS (ML Performance)				
Precision (1 – FPR)	TBD baseline 0.51 @ threshold 0.7	≥ 0.95	Confusion matrix, monthly model eval	Primary quality gate. Directly controls false positive rate experienced by analysts.
Recall (True Positive Rate)	TBD baseline	≥ 0.85	Confusion matrix, monthly model eval	Secondary constraint. Catch $\geq 85\%$ of actual fraud events.

	0.87 @ threshold 0.7			
F1 Score	TBD baseline	≥ 0.90	Harmonic mean of Precision & Recall	<i>Primary model eval metric; balances precision/recall tradeoff.</i>
AUC-ROC	TBD baseline	≥ 0.92	Sklearn roc_auc_score on held-out test set	<i>Threshold-independent signal; measures model's ability to rank fraud above legit across all cutoffs.</i>
SHAP Explanation Coverage	N/A	$\geq 95\%$ of alerts	Count of alerts with valid SHAP output / total alerts fired	<i>Tracks explainability layer reliability — flags infra failures silently missing feature attribution.</i>
Model Drift (PSI)	N/A	PSI < 0.10 per monthly run	Population Stability Index on score distribution vs. training baseline	<i>Early warning for data/model drift before it degrades Precision/Recall metrics.</i>
Model Retraining Cycle	Ad hoc	Monthly automated	Pipeline execution logs (Airflow/cron)	<i>Ensures model stays calibrated to evolving fraud patterns via analyst feedback labels.</i>

LAYER 2 — PRODUCT METRICS (User & Business Outcomes)

Alert-to-Decision Time	~8 min/alert	≤ 5 min/alert	Analyst dashboard timestamps (decision log)	<i>Core workflow efficiency metric. Reduction validates that LLM explanations + UI design are working.</i>
Analyst Feedback Rate	N/A (new)	$\geq 80\%$ of alerts actioned	% alerts receiving Confirm/Reject/Escalate click within SLA window	<i>Adoption signal. Low rate = analysts ignoring the queue or not trusting the system.</i>
Analyst Escalation Rate	~40% of alerts	$\leq 28\%$ of alerts	Alert outcome tracking (escalate vs. self-resolved)	<i>Proxy for explanation quality. High escalation = analysts can't make decisions without extra help.</i>
LLM Explanation Latency	N/A	≤ 1 second	API response time monitoring (p95)	<i>UX SLA. Latency above 2s breaks analyst flow and degrades perceived system intelligence.</i>
LLM Explanation Usefulness	N/A (new)	$\geq 70\%$ thumbs-up	In-queue 👍/👎 rating on each explanation (optional analyst input)	<i>Qualitative signal that NLP layer adds value vs. noise. Feeds prompt iteration.</i>
False Negative Cost (Est.)	N/A (new)	Track; alert if > ₹X/month threshold	Estimated financial loss from undetected fraud (FN count × avg fraud value)	<i>Business-level accountability metric. Ensures Precision optimisation doesn't silently increase missed-fraud cost.</i>

False Positive Rate	~15% estimated	≤ 5%	Analyst feedback labels (Mark Legitimate actions)	<i>Direct customer experience metric. Each FP is a blocked legitimate transaction.</i>
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Key design decision: FPR and Precision are the same metric restated — only Precision is tracked to avoid double-counting. AUC-ROC is added as a threshold-independent model health signal. Analyst Feedback Rate and LLM Explanation Usefulness are new product adoption metrics not in v0. False Negative Cost is tracked as a business accountability metric to ensure Precision optimization does not silently increase missed-fraud exposure.

9. Risk Assessment & Mitigation

Risk	Category	Likelihood	Impact	Mitigation
Class imbalance in training data	Technical / ML	High	High	Apply SMOTE; monitor precision/recall separately; use F1 as primary model eval metric
LLM hallucination in explanations	Technical / AI	Medium	High	Ground prompts strictly in transaction metadata; add output validation layer; fallback to rule-based explanation if LLM fails
Model drift over time	Technical / ML	High	Medium	Automate monthly retraining pipeline using analyst feedback labels; track PSI degradation alerts
High false positive rate alienates analysts	Product / UX	Medium	High	Threshold tuning during beta; analyst feedback loop; configurable sensitivity slider for team leads
Regulatory compliance (GDPR / RBI)	Compliance	Medium	High	Ensure all flagged data is anonymized; maintain immutable audit log; legal review before production
Analyst over-reliance on AI	Operational	Low	Medium	Display confidence intervals prominently; require analyst note for high-value escalations; periodic model performance reviews

10. Technical Stack Overview

Layer	Technology	Purpose
Training Data	European Credit Card Fraud Dataset (Kaggle)	284,807 transactions, 492 fraud cases
Classification Model	XGBoost / Random Forest	Binary fraud classification, SHAP explainability

Imbalance Handling	SMOTE (imbalanced-learn)	Oversample minority class during training
Explainability	SHAP + LLM (Claude / GPT-4o)	Feature importance + natural language explanation
Backend API	Python / FastAPI	Real-time scoring endpoint (< 300ms)
Frontend	Streamlit (v1) → React + Tailwind CSS (v2)	Analyst alert queue dashboard
Database	PostgreSQL	Transaction records, alert decisions, audit log
Pipeline	Apache Airflow (or cron)	Monthly model retraining automation

11. Out of Scope (v1.0)

- Customer-facing fraud notification flow (post-detection communications)
- Chargeback management or dispute resolution workflows
- Multi-currency or cross-border transaction support
- Native mobile app for analysts (web-responsive UI only)
- Real-time streaming data pipeline (batch processing sufficient for v1.0)
- Integration with third-party fraud intelligence feeds (e.g., Threatmetrix)

12. Open Questions

- What is the acceptable threshold for the risk score before an alert is surfaced? (Default: 70 — requires calibration with fraud ops team)
- Should the system support custom model thresholds per merchant category or user segment?
- What is the SLA for fraud resolution — does the system need to surface alerts within a specific time window?
- Who owns the LLM prompt library — PM, Data Science, or Compliance?
- Does the compliance team require model explainability documentation for regulatory audits?

Appendix — Confusion Matrix Reference

Understanding the confusion matrix is central to FraudShield Agent's success metrics:

	Predicted: Fraud	Predicted: Legit
Actual: Fraud	True Positive (TP) Caught fraud ✓	False Negative (FN) Missed fraud ✗
Actual: Legit	False Positive (FP) Wrong alert — minimize!	True Negative (TN) Correctly ignored ✓

Precision = $TP / (TP + FP)$ — Of all alerts raised, how many were real fraud?

Recall = $TP / (TP + FN)$ — Of all actual fraud, how many did we catch?

F1 Score = $2 \times (Precision \times Recall) / (Precision + Recall)$ — Harmonic mean; primary model eval metric

AUC-ROC — Threshold-independent measure of model discrimination ability (higher = better across all cutoffs)

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